

Section 14.1: Functions of Several Variables

Partial Derivatives

MTH 310
Calculus III

Functions of Several Variables

In Calc I and II, every function had **one input** and **one output**: $y = f(x)$.

But in the real world, most quantities depend on **more than one variable**:

- Temperature in a room depends on position (x, y, z)
- Your grade depends on homework, quizzes, and exams
- Wind chill depends on temperature *and* wind speed

Definition 1: Function of Two Variables

A **function of two variables** is a rule that assigns to each ordered pair (x, y) in a set $D \subseteq \mathbb{R}^2$ a unique real number z :

$$z = f(x, y)$$

- **Domain**: the set D of valid inputs (x, y)
- **Range**: the set of all output values $z = f(x, y)$

This extends naturally: $w = f(x, y, z)$ is a function of **three** variables, and so on.

Same idea as Calc I, but now the input is a **point** in $\mathbb{R}^2$ (or $\mathbb{R}^3$) instead of a single number.

Example 1: Finding the Domain

Example 1

Find and sketch the domain of $f(x, y) = x \ln(x^2 - y)$.

Look for restrictions, set up and solve inequalities, sketch.

Your Turn: Find the Domain

Example 2

Find the domain of $g(x, y) = \frac{\sqrt{y - x^2}}{1 - x^2}$.

Identify both restrictions, then describe and sketch the domain.

Graphs of Functions of Two Variables

In Calc I, the graph of $y = f(x)$ was a **curve** in $\mathbb{R}^2$.

Now the graph of $z = f(x, y)$ is a **surface** in $\mathbb{R}^3$:

Definition 2: Graph of a Function of Two Variables

The **graph** of f is the set of all points (x, y, z) in $\mathbb{R}^3$ such that $z = f(x, y)$ and $(x, y) \in D$.

$$\text{graph} = \{(x, y, z) \mid z = f(x, y), (x, y) \in D\}$$

Input = point in the xy -plane. Output z = height above (or below) that point. The collection of all these heights forms a **surface**.

Example 3

Sketch the graph of $f(x, y) = -3x - 2y + 6$.

Recognize: This is $z = -3x - 2y + 6$, which is **linear** in x and y . A linear equation in three variables is a **plane**.

Find the intercepts and sketch the plane.

Traces: Slicing a Surface

A 3D surface can be hard to visualize all at once. One strategy: **slice it** with a plane and look at the resulting curve.

Definition 3: Traces

A **trace** of a surface is the curve of intersection when the surface is sliced by a plane.

- **Horizontal trace** (or *cross-section*): set $z = c$ (a constant) and see what curve remains in the xy -plane
- **Vertical trace**: set $x = c$ or $y = c$ and see the resulting curve

Analogy: Think of slicing a loaf of bread. Each slice reveals a 2D cross-section of the 3D shape.

Or think of a topographic map — each contour line is where the terrain has a specific elevation.

Traces reduce a 3D problem to a **2D curve** — something you already know how to analyze from Calc I.

Level Curves and Contour Maps

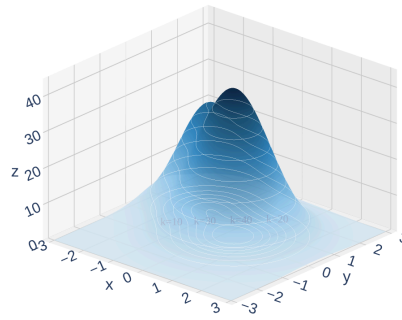
Definition 4: Level Curves

The **level curves** of $f(x, y)$ are the horizontal traces $f(x, y) = k$ for various constants k .

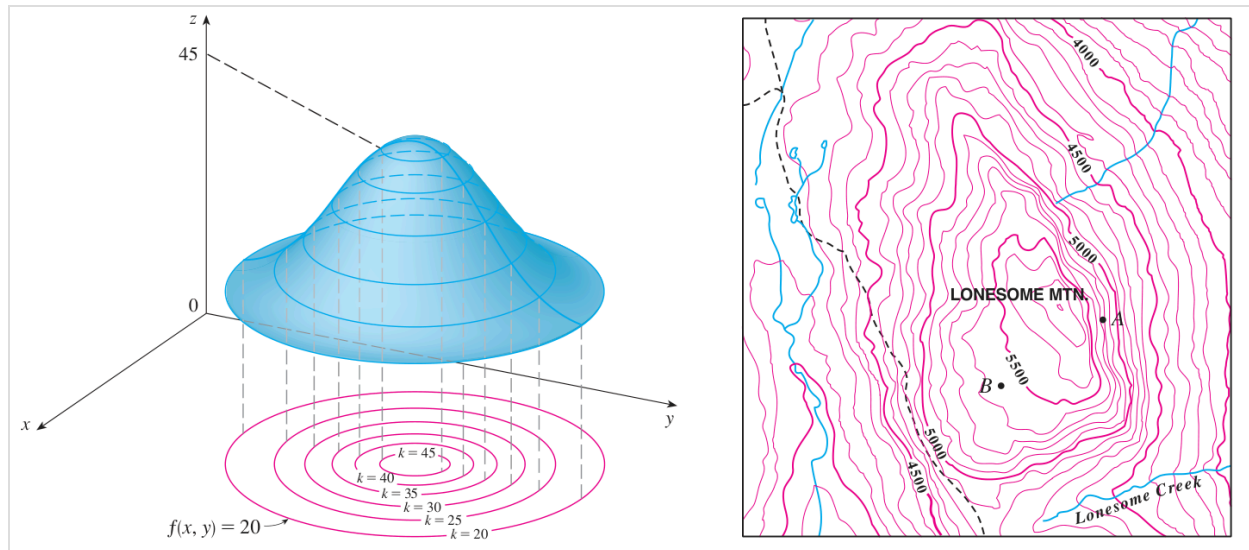
Each level curve is drawn in the xy -plane. Viewed together, they form a **contour map** (or *contour plot*).

You've seen these before!

- Topographic maps: curves of constant elevation
- Weather maps: curves of constant temperature (isotherms) or constant pressure (isobars)
- Closely spaced contour lines = the surface is **steep**
- Widely spaced = the surface is **gentle**



Drag to rotate — see how the surface produces the contour map on the floor | [Desmos 3D](https://www.desmos.com/3d) [https://www.desmos.com/3d]



Contour lines show constant elevation | [Desmos 3D](https://www.desmos.com/3d) [https://www.desmos.com/3d]

A contour map compresses 3D information into a 2D picture. Reading one is like reading a topographic map — each curve is a constant height $z = k$.

Example 4: Contour Plot of $f(x, y) = x^2 - y^2$

Example 4

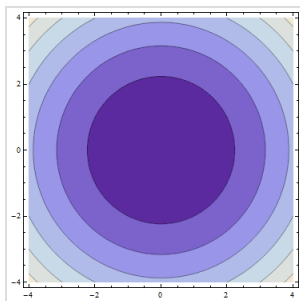
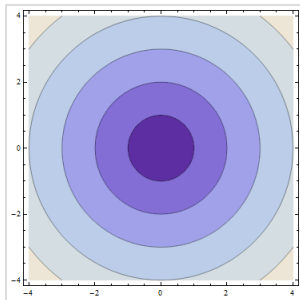
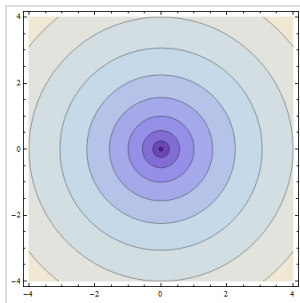
Sketch the level curves of $f(x, y) = x^2 - y^2$ for $k = -4, -1, 0, 1, 4$.

Match the Contour Plot to the Surface

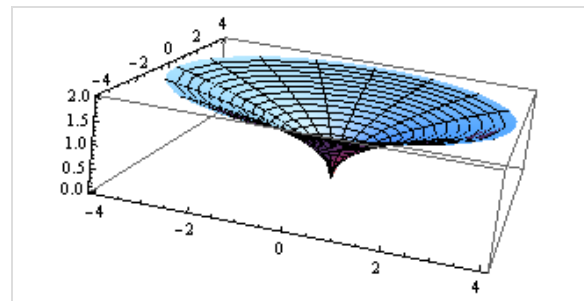
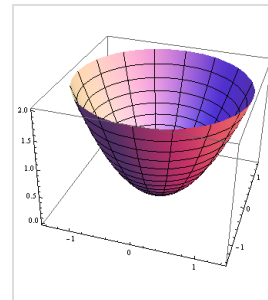
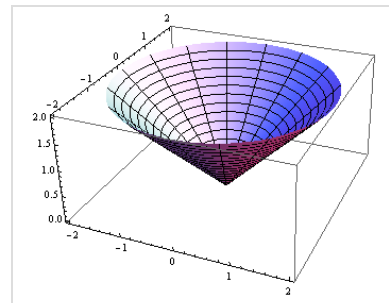
Challenge: Can you match each contour plot (left) to its surface (right)?

Look for clues: symmetry, shape of level curves, spacing of contour lines.

Contour Plots



Surfaces



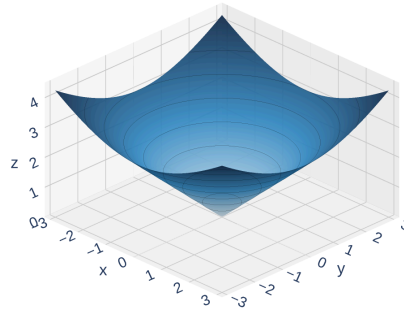
Write your matches and explain your reasoning.

The Equations Behind the Surfaces

All three surfaces have the form $z = r^p$ where $r = \sqrt{x^2 + y^2}$. The **power** p controls the contour spacing. Rotate to look straight down to see the contour map!

Cone: $z = \sqrt{x^2 + y^2}$ — constant slope, so contours are **evenly spaced**.

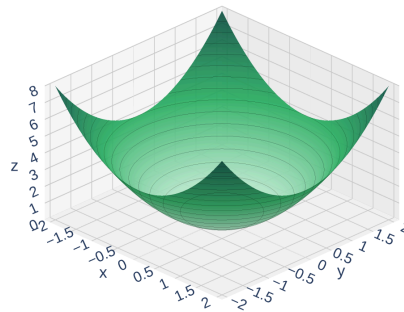
$$z = \sqrt{(x^2 + y^2)}$$



[Desmos 3D](https://www.desmos.com/3d) [https://www.desmos.com/3d]

Paraboloid: $z = x^2 + y^2$ — slope increases outward, so contours get **closer together**.

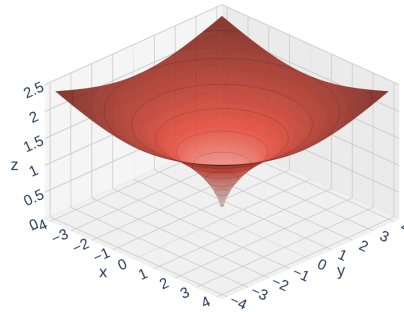
$$z = x^2 + y^2$$



[Desmos 3D](https://www.desmos.com/3d) [https://www.desmos.com/3d]

Fourth root: $z = (x^2 + y^2)^{1/4}$ — slope decreases outward, so contours get **farther apart**.

$$z = (x^2 + y^2)^{1/4}$$



Desmos 3D [<https://www.desmos.com/3d>]

Level Surfaces: Functions of Three Variables

What if the function has **three** input variables, like $w = f(x, y, z)$?

We can't graph $w = f(x, y, z)$ directly — that would require **4 dimensions**!

Instead, we visualize using **level surfaces**: set $f(x, y, z) = k$ for various constants k .

Each value of k gives a **surface** in $\mathbb{R}^3$ (not just a curve).

Definition 5: Level Surfaces

The **level surfaces** of $f(x, y, z)$ are the surfaces defined by $f(x, y, z) = k$ for constants k in the range of f .

Analogy:

- For $f(x, y)$: level *curves* (1D objects in 2D space)
- For $f(x, y, z)$: level *surfaces* (2D objects in 3D space)
- Same idea, one dimension up!

Level curves collapse a 3D graph to 2D. Level surfaces collapse a 4D graph to 3D. Both let us visualize functions we can't directly graph.

Example 6: Level Surfaces of a Temperature Field

Example 6

A temperature distribution is given by:

$$T(x, y, z) = \frac{100}{(x^2 + y^2 + z^2)^{3/2}}$$

Describe the level surfaces. What shape are the level surfaces? What happens to temperature as you move away from the origin?

Set $T = k$ and solve for the relationship between x , y , and z .

Homework

Section 14.1: 3, 9, 13, 23-31, 32, 36, 41, 42, 45, 47, 49, 61-66, 67, 68